Gradient – Data Path

Equipment Room

Main Disconnect Panel CB3 TB1-15-16 CB2
Remote MRU
E-Off Switch
O_2 Monitor Control Unit: Meter, Ctrl Board, Alarm, Light
Customer-Supplied Outlet
Site E-Net
Facility Power Filters
Facility Utilities: Facility Coolant Supply, Facility Coolant Return

Mag Mon J5 J10 J8 J9 J7
HEC: HEC Power Box (Cryo, Power Supply, GC, PE, Blower); Signal Box J24 J11 J12; Grad HtXchr; Grad Res; Grad Pump; Flow Meter; Filter; Blower HtXchgr; PGR Pump; PGR Res; PGR HtXchr; Grad/RF Coil Blower
JR Power Cryo-Cooler Compressor JC Water Return, Water Supply, Gas Return, Gas Supply

PGR
PDM J29 J25 L1/L2/L3 J27 J28 J24
VRE 2 (32-ch) J5 Net 3 Net 2 Net Mgmt J1
VRE 1 J1 Net Mgmt Net 2 Net 3 J5
ETH 1 2 3 4 Term Server
Aux Board J4 J6 J1 J2/J3 J9 J11 J10
XRFD LS1-A LS1-B LS2-A LS2-B LS3-A LS3-B LS4-A Term; PDU Top 1; PDU Top 1; PDU Front
J4 XGD Fan x3
Unused
CB1 J1 J4 J3 PGR PDU Host J6/J7/J8 J11/J12/J13 Recon RF/MNS J5 J2 PEN
CB2 J14 NB Amp J9 J10 J20 J5 J11 J7 J15 J4 J3 J18 J1 J2 J19 J16 J8 J12 J6
16 Port E-net Sw
J3 J6 J2 J102 J19 J18 J10 J9 J17 J14 J15 J13 J12 J4
Manifold Supply: SMA Coax
Pwr XPS x3 J2 J10 J1 J11 J8 J9 J3 J4/J5 J4/J5
XGA x3 J8 J7 J16 J13 J15 J14 J5 J12 J6 J17 J18 J1/J2 J1/J2
CAM: NB AIF J8 J3; UPM1 Det J1 J2 J4 J5 J3 Proc J3; UPM2 J1 Det J2 J4 J5 J3 Proc J3; SCP2 ETH J2 PMC J3; 10/100 Base AGP2; SRF/TRF STIF J4 J3 J18/J19; IRF3 J23 I/O Panel J10 J8 J14 J13 J12 J11
Cronus Chassis: XPS CB J13/J16 J12/J15 J11/J14 P1; XGA CB J13/J16 J12/J15 J11/J14; J10/J11 J6 J9 GMC J7 J8; Cronus Chassis Optical Fiber Communication; J3 J4 J5; 6 6 6; J2 J1
1 2 3 4 5 6
J8 J20 J204 J201 J200 J203 J202 J205 J22 J11 J21 J1 J31 J5 J16
Unused
Path Info
The signal path for the boards in the CAM Chassis passes through the boards in the following order:
1. Waveform commands originate in the AGP and are passed on to the SRF/TRF.
2. From the SRF/TRF to the IRF3, via the backplane,
3. From the IRF3 to the Cronus Chassis, GMC board.
4,5. The clock signal originates in the Exciter and is passed to the Cronus Chassis GMC board, and CAM IRF3 board for synchronization.
6. Gradient Pulse and clock signals are passed from the GMC to the XGAs for pulse creation.

PEN J14 J1
NB Exciter J19 J20 J21 J18 J7 J23 J22 J12 J29 J32 J30 J15 J14 J17 J1
Power Distribution J1 J4 J2 J3
PHPS J12 J9 J37/J38 J2 J43 J39 J54 J41 J40 J42 J35 J36 J44 J8 J11 J1 J5 J10
CFB J18 J19 J7/J8 J1/J2 J5/J6 J3/J4 J17
Driver Module J14 J8 J19 J17 J11 J20 J12 J13 J1 J2 J7 J6 J23 J25 J21
SRPS J6 J5 J1 J51 J52 J46 J48 J45 J50 J47 J3
DD Filter J94 J96 J92 J74 J76 J72
J4 J3 J9 J5 J6 J16 J39 J54 J41 J40 J42 J35 J36 J51 J52 J46 J48 J45 J50 J47 J29 J32 J30 J34 F11 F10 F9 F8 F7 F6 F5 J31 J33

Door Switch
Coolant Removal Kit
Temp/Hum Sensor
Scan Room Blower Box
Patient Comfort
Hub

Data Path – General
The Host PC is connected to the Gradient chain through the Ethernet Switch. The Gradient information, in turn, passes through the Ethernet Switch to the CAM Chassis via the AGP Board. From the AGP board, information is passed to the SRF/TRF board via the CAM backplane. The SRF/TRF board operates as the brains of the process, passing the signal on to the other two boards via the CAM backplane.

Once fully processed the signal is sent to the Cronus Chassis in the XGD before the Gradient Pulse is generated.

SPW J115 J116 J117 J16 J106 J108 J107 J118 J17
Term Conn F1 F2 F3 F4
Gradient Filter X Y Z
J121 J122 Xmit Sw1 J120 J123
104 Wvgde 105 Wvgde
J100 Wvgde J101 Wvgde J111 Wvgde J110 Wvgde Patient Alert Wvgde
Primary MRU
E-Off Switch
O_2 Monitor Sensor Alarm Light
Mag Enclosure Chimney
Gradient Bus Bar X Y Z
Mag I/O J1 J3 J6 J4 J2 J5 J14 J15 J12 J11 J13 J8 J7 J9 J10 J17 J16
FO I/F J28 J20 J21

Operator Workspace

Temp/Hum Sensor
Server (Option)
Service Laptop
Optional Speaker
GOC: IRDC (Option) J5 J4 J3 J2 J1; USB-Serial Converter; Ethernet Switch 1 2; GOCAA J12 J14 J17 J18 J15 J7 J6; Host PC Ethernet B Lwr Graphic Card Port 1 USB RS232 USB USB USB Upr Graphic Card Port 1 Ethernet A Line In Line Out; J10 J4 J6 J9 J2 GOC J5 J1 PDU J3
Global Modem (Option)
Key-board
Mouse
Primary Display
Pneumatic Patient Alert System

Scan Room

Mag Bore: J6 J7 Body Coil J2 J3 J9; Bore-PAL L R; Left Arc Cover; Ambient Temp Snsr; Bore Temp Sensors; Front Mic; Airflow Sensors; XRMB X Y Z Gradient Temp Sensor; J101 MAG-Grad Coil; J100 MAG-Grad Coil; J104 Grad Coil Air; Patient Air
Magnet: J3 P403 Terminal Box; Pressure Transducer; Reed Switch; J2 RUO PreAmp J1; Coldhead Pwr Sleeve Coldhead Supply Return; Airflow Sensor Module; J9/J10 J2 PAC; SRI4 J23 J10 J34 J3 J20 J32 J24 J17 J1 J4 J5 J11/J12 J13/J14 J33 J21 J31 J2 J22; PreAmp & Combiner J11 J10 J9; J1 J5 J2 J6 J3 J7 BHYB J4; LCP J11 J3; RCP J11 J3; IRD J1; J11 J2 J9 J7 J10 TDM J1 J12; J9 RRx1 J7 J1 J4/J5 J8 J2; J7 RRx2 (32-ch) J1 J4/J5 J2
MAG-PAC J4
PAC I/F
Squeeze Bulb
Speaker
Left E-Stop Button
Rt E-Stop Button

Rear Pedestal: J14 J13 Dummy Load J12HYB; PED Antenna; Lights
RF Hub: Hub Backplane (bias, power, status/control); J102 J101 J104 J103; J1 J3 J2 J4 PDB J5 J6; MCDB; J9 J13 J12 J16 J17 RFCB J10 J8 J14/J15 J7 J6 J1 J2 J3 J4; Debug Port; J1–J6 (ch 17-32); RFSB (ch 1-16) J1–J6
J4 J6 J3 TIF J1 J2 J5
Long Drive J1 ASM (MTR) J2
POS Enc
J3 EOT Sensor HOME Sensor EGRESS Sensor
LPCA: J4 J2 Port A J5 J1; Port P2 J4; Port P1 J5; Rear Mic
Load A
Load B
Reroute Box: J10 J7 P2A P2B P1A P1B J8 P3A P3B P4A P4B J9
J1 PED I/O J2
Rear Pedestal Air

Patient Table: Coil Port P3; Coil Port P4; Touch N Go_R; Touch N Go_L; Table Hydraulics
Dock: J2 Mag J1 Dock J3 Left; ODU

Key

TBD = To Be Determined (Future Enhancement)
Cabinet
Chassis/ Sub-system
Part
Data Cables
Coolant
Power
Pulse
Data Paths

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